

sources_1\new\RESET_SYNC.vhd

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  --*****
5  --*
6  --* Name: RESET_SYNC
7  --* Designer: Matthew Mwangi
8  --*
9  --* This is a component that handles synching its input in this case the reset
10 --* active clock, to avoid metastability issues.
11 --*
12 --*****
13
14
15 entity RESET_SYNC is
16   Port (
17     reset: in std_logic;
18     clock: in std_logic;
19     synchedReset: out std_logic
20   );
21 end RESET_SYNC;
22
23 architecture RESET_SYNC_ARCH of RESET_SYNC is
24   constant ACTIVE_RESET: std_logic := '1';
25   signal chain: std_logic_vector(1 downto 0);
26 begin
27   process(reset, clock)
28   begin
29     if reset = ACTIVE_RESET then
30       chain <= (others=>ACTIVE_RESET);
31     elsif rising_edge(clock) then
32       chain <= chain(0) & not ACTIVE_RESET;
33     end if;
34   end process;
35
36   synchedReset <= chain(chain'high);
37 end RESET_SYNC_ARCH;
38
```